

Homework - seminars 3, 4, 5 and 6

1. Consider $f: \mathcal{P}(n) \rightarrow \mathbb{N}$ $p \in \mathcal{P}(n), f(p) = |\{(i, j) / i < j, p(i) = j \text{ \textit{ve}} p(j) = i\}|$ the objective function of a maximization problem, where $\mathcal{P}(n)$ denotes the set of n-sized permutation.

a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.

b. Give a mutation probability (*pm*), write a mutation function using the insert mutation operator, that computes a new population (*popm*) from the population *pop*. The size of the new population is also *dim*.

2. Consider $f: \{1, 2, \dots, 1500\} \times \{-1, 0, \dots, 2500\} \rightarrow \mathbb{R}, f(x, y) = y * (\sin(x - 2))^2$ the objective function of a maximization problem. For each phenotype $(x, y) \in \{1, 2, \dots, 1500\} \times \{-1, 0, \dots, 2500\}$, the corresponding genotype is a binary string that concatenates the standard binary representation of the two phenotype components.

a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.

b. Given a crossover probability (*pr*), write a crossover function using the multi-point crossover operator with 3 crossover points, that creates a new population (*popc*) from the population *pop*. The size of the new population is also *dim* (asexual crossover is used when needed). The fitness of each chromosome is also stored at the end of the chromosome representation.

3. Consider $f: [-1, 1] \times [0, 0.2] \times [0, 1] \times [0, 5] \rightarrow \mathbb{R}, f(x_1, x_2, x_3) = 1 + \sin(2x_1 - x_3) + (x_2 * x_4)^{1/3}$ the objective function of a maximization problem. A genotype is a vector $x = (x_1, x_2, x_3, x_4)^T, x \in [-1, 1] \times [0, 0.2] \times [0, 1] \times [0, 5]$

a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.

b. Give a mutation probability (*pm*), write a mutation function using the creep mutation, with the magnitude $t = 0.6 \left(\sigma = \frac{t}{3} \right)$, that creates a new population (*popm*) from the population *pop*.

4. Consider $f: [-1, 1] \times [0, 1] \times [-2, 1] \rightarrow \mathbb{R}, f(x_1, x_2, x_3) = 1 + \sin(2x_1 - x_3) + \cos(x_2)$ the objective function of a maximization problem. A genotype is a vector $x = (x_1, x_2, x_3)^T, x \in [-1, 1] \times [0, 1] \times [-2, 1]$

a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.

b. Given a crossover probability (*pr*), write a crossover function using the total arithmetic crossover operator, that creates a new population *popc* from the population *pop*. The size of the new population is also *dim* (asexual crossover is used when needed) and the fitness of each chromosome is also stored at the end of the chromosome representation.

5. Consider the function $f(x) = \sum_{i=1}^7 x_i$, $x = (x_1, \dots, x_7) \in \{0,1\}^7$ that must be maximized (a genotype is a binary vector with 7 components).

a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.

b. Given a crossover probability (*pr*), write a crossover function using the multiple-point crossover operator with 2 crossover points, that creates a new population (*popc*) from the population *pop*. The size of the new population is also *dim* (asexual crossover is used when needed) and the fitness of each chromosome is also stored at the end of the chromosome representation.

6. Consider $f: \{1,2, \dots, 350\} \rightarrow \mathbb{R}$, $f(x) = x^2$ the objective function of a maximization problem. For each phenotype $x \in \{1,2, \dots, 350\}$, the corresponding genotype is a Gray representation binary array.

a. Write a Python function that generates a random population (*pop*), with size *dim*.

b. Given a crossover probability (*pr*), write a crossover function using the single-point crossover operator, that creates a new population (*popc*) from the population *pop*. The size of the resulting population is also *dim* (asexual crossover is used when needed).

7. Consider $f: \mathcal{P}(n) \rightarrow \mathbb{N}$ $p \in \mathcal{P}(n)$, $f(p) = |\{(i,j)/i < j, p(i) = j \text{ \textit{and} } p(j) = i\}|$ the objective function of a maximization problem, where $\mathcal{P}(n)$ denotes the set of n-sized permutation.

a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.

b. Give a mutation probability (*pm*), write a mutation function using the scramble mutation operator, that creates a new population (*popm*) from the population *pop*. The size of the resulting population is also *dim*.

8. Consider $f: \mathcal{P}(n) \rightarrow \mathbb{N}$ the objective function defined for the n-queens problem as: $p \in \mathcal{P}(n), f(p) = n \times \frac{n-1}{2} - |\{(i, j) / i < j, |p(i) - p(j)| = |i - j|\}|$, where $\mathcal{P}(n)$ denotes the set of n-sized permutation.

- a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.
- b. Use the function implemented for point a) to generate two populations (*pop1* and *pop2*), with *dim* individuals each. Write a Python function that creates a new population by applying the elitist selection on the previously generated populations, considering *pop1* as current generation and *pop2* as descendants. The size of the resulting population is also *dim*.

9. Consider $f: \{1, 2, \dots, 2500\} \rightarrow \mathbb{R}, f(x) = (\sin(x - 2))^2$ the objective function of a maximization problem. For each phenotype $x \in \{1, 2, \dots, 2500\}$, the corresponding genotype is the standard binary representation of x .

- a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.
- b. Write a Python function that selects a mating pool from the population *pop* using the roulette mechanism with sigma-scaled FPS probability distribution.

10. Consider $f: \{1, 2, \dots, 350\} \rightarrow \mathbb{R}, f(x) = x^2$ the objective function of a maximization problem. Fiecărui fenotip $x \in \{1, 2, \dots, 350\}$ îi corespunde un genotip șir binar obținut prin codificarea Gray.

- a. Write a Python function that generates a random population (*pop*), with size *dim*.
- b. Use the function implemented for point a) to generate two populations (*pop1* and *pop2*), with *dim* individuals each. Write a Python function that creates a new population by applying the GENITOR mechanism (replacing 2 individuals), considering *pop1* as current generation and *pop2* as descendants. The size of the resulting population is also *dim*.

11. Consider $f: \{1, 2, \dots, 500\} \rightarrow \mathbb{R}, f(x) = (\sin(x - 2))^2 - x \cdot \cos(2 \cdot x)$ the objective function of a maximization problem. Fiecărui fenotip $x \in \{1, 2, \dots, 500\}$ îi corespunde un genotip șir binar obținut prin reprezentarea standard în bază 2 a lui x .

- a. Write a Python function that generates a random population (*pop*), with size *dim*; the fitness of each individual is stored at the end of each chromosome representation.
- b. Write a Python function that selects a mating pool from the population *pop* using the tournament mechanism with k participants (k is an input parameter).